Assessment Plan for PHY4605 (Semester 1 2026/2027)

Course code and name: Kaedah Pengkomputeran dalam Fizik / Computational Methods in Physics (PHY4605)
Course type: ~~1. General / Basic~~ ~~2. Fully Practical~~ 3. Focused
Credits: 3 (2+1)
Total Student Learning Time: 120 hours
Cohort: ~~Year 1~~ ~~Year 2~~ Year 3 Year 4
Lecturer: Dr Muhammad Khairul Adib Muhammad Yusof
Ts. Dr. Muhammad Kashfi Shabdin

Course type	Cognitive	Psychomotor	Affective
General course	60–80%	5–40%	5–40%
Focused course	60–80%	5–40%	5–40%
Practical / Fully Practical course	30–60%	30–60%	≥5%

Activity	Type of Assessment	Week	Programme Outcome Contribution Course-weight allocation (%)										Total (%)	Remarks
			PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10		
Practical portfolio	Formative	2–11		15									15	Ten packets; best 8 counted; 12% group + 3% individual defence
Individual lab test	Summative	TBC		10									10	Scaffolded verification + parameter/defect variant + defence
Capstone investigation	Formative	13–14								5	10		15	Bounded group; checkpoints 06–12; presentation, defence, handoff
Test 1	Summative	6	10										10	90 min open-book; Weeks 01–05
Test 2	Summative	12	10										10	90 min open-book; Weeks 06–11
Final examination	Summative	16–17	40										40	120 min cumulative open-book
Grand total			**60**	**25**	**0**	**0**	**0**	**0**	**0**	**5**	**10**	**0**	**100**	

Mapping CO-PO for PHY4605

CO	Hasil Pembelajaran Kursus (CO)	Hasil Pembelajaran Program (PO)			
		PO1 C	PO2 PP	PO8 KK	PO9 LS
1	Menaksir kaedah analisis berangka berkaitan fizik (C5, LS)	✓			✓
2	Mengorganisasi pelbagai teknik analisis berangka dalam penyelesaian masalah fizik (P4)		✓		
3	Mengenal pasti potensi kaedah pengkomputeran untuk tujuan pengkomersilan (A4, KK)			✓	
	Jumlah (%)	**60**	**25**	**5**	**10**

Kod hasil pembelajaran program

PO1 – Pengetahuan (C)
PO2 – Kemahiran Teknikal / Praktikal / Psikomotor (PP)
PO3 – Pendekatan Kemahiran berfikir dan saintifik (CTPS)
PO4 – Kemahiran berkomunikasi (CS)
PO5 – Kemahiran sosial dan bertanggung-jawab (TS)
PO6 – Profesionalisme, nilai, sikap, etika dan moral (EM)
PO7 – Pendidikan sepanjang hayat dan pengurusan informasi (LL)
PO8 – Kemahiran personal dan keusahawanan (KK)
PO9 – Kemahiran kepimpinan, autonomi dan tanggungjawab (LS)
PO10 – Kemahiran Numerasi (NS)